Cannabis vape cartridge fill (vape oil): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for cannabis vape cartridge fill — the consumable vape oil, sometimes called 'juice' — evaluated on the finished inhalable consumable as sold. Each limit is the strictest sovereign cannabis heavy-metal action level applicable to the inhalation route (the same pool as flower); total chromium is a screening ceiling with hexavalent chromium as the binding control, and inorganic arsenic, methylmercury and Cr(VI) are enforced by reflex speciation. This standard covers the cartridge fill only; the vape hardware and its aerosol emissions are a separate reserved determination.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default, route-preserving; strictest applicable sovereign cannabis maximum; occurrence never selects a value)

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	500	sovereign maximum level
Cadmium (Cd)	200	sovereign maximum level

Total arsenic (tAs)	200	sovereign maximum level
Inorganic arsenic (iAs)	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	100	sovereign maximum level
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	800	human-grade read-across (European Union)
Tin (Sn)	250,000	human-grade read-across (Singapore)
Aluminium (Al)	100,000	human-grade read-across (China)
Total chromium (Cr)	600	sovereign maximum level
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	500	government maximum level
Cadmium (Cd)	200	government maximum level
Total arsenic (tAs)	200	government maximum level
Inorganic arsenic (iAs)	≤ tAs	iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell.
Mercury, total (tHg)	100	government maximum level
Methylmercury (MeHg)	≤ tHg	MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell.
Nickel (Ni)	800	European Union — human-food analog
Tin (Sn)	250,000	Singapore — human-food analog
Aluminium (Al)	100,000	China — human-food analog

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Total chromium (Cr)	600	government maximum level
Hexavalent chromium (Cr-VI)	≤ Cr	Cr-VI carries no product-specific sovereign cell; it is a component of and bounded by its parent total Cr (Cr-VI <= Cr), an enforceable ceiling via reflex speciation rather than an empty cell.

4 | Cite this standard

Pendergrass, K. (2026). *Cannabis vape cartridge fill (vape oil): HMTc standard* (Standard No. 10.83138/hmtc.std.25.cannabis-vape-cartridge-fill). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.25.cannabis-vape-cartridge-fill

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References

[1] China — human-food maximum levels
Binds Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] Singapore — human-food maximum levels
Binds Sn. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/cannabis-vape-cartridge-fill. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

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